

Roberto Olvera Hernández

Bioinformatics Scientist

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in [rob-olvhez](#) 🔗 [robolvhez](#)

Bioinformatics Scientist specializing in computational genomics and cloud infrastructure. I manage terabyte-scale genetic data for the Mexico City Prospective Study in collaboration with Oxford University. I build automated ETL pipelines on DNAnexus and develop web platforms to make complex genomic data accessible. My primary focus is writing scalable code to untangle disease risks for diverse, mixed-ancestry populations that are often left out of standard research.

📁 Professional & Research Experience



Computational Genomics Researcher

Centro de Ciencias Genómicas (CCG) — Universidad Nacional Autónoma de México (UNAM)

Feb. 2025 - Present

Cuernavaca, MX

Leading design and implementation of a scalable pipeline for genome-wide association studies (GWAS) in the Mexico City Prospective Study (MCPS), a cohort of over 140,000 patients, using cloud computing infrastructure.

- **Software Development:** Engineered a scalable ETL pipeline (Python, R, and Bash) to process large-scale clinical baseline and genotype data to perform within-family GWAS and downstream analyses (e.g., LD scores, heritability, and variant-level comparisons).
- **Cloud Infrastructure:** Maximized the AWS-based DNAnexus platform to automate our ETL pipeline, dynamically provisioning compute resources to optimize execution times and significantly reduce operational costs.
- **Data Standardization:** Identified that unstandardized data was blocking cross-cohort research. Built a prototype ETL pipeline to map over 200 clinical phenotypes into the OMOP model to solve this interoperability problem.
- **Cross-functional Collaboration:** Partnered with international teams to ensure production-readiness and strict data governance for large-scale genetic pipelines.
- **AI-Assisted Engineering:** Leveraged AI coding tools (Antigravity CLI) for rapid prototyping, pipeline testing, and automation of complex data workflows.



Genetic Epidemiology — Research Placement

Nuffield Department of Public Health (NDPH) — University of Oxford

Aug. 2025 - Jan. 2026

Oxford, U.K.

Invited as a visiting researcher to work directly with the Oxford team managing the genomic cohort. I used this placement to deeply understand the data structure and build strong relationships with NDPH researchers.

- **Workflow Standardization:** Collaborated with the team to design a standard analysis workflow for 16 complex traits. I later scaled this framework to cover additional clinical traits relevant to the Mexican population.
- **Cloud Automation:** Trained directly on the DNAnexus platform. I applied this experience to architect and fully automate our cloud-based ETL pipelines.
- **Genetic Epidemiology:** Sharpened my analytical skills by working alongside senior researchers to evaluate complex genomic data.



Income Data Analyst

TOKA Internacional

2024 - 2025

Zapopan, MX

Spearheaded the modernization of a fragile billing process for millions of monthly transactions, navigating a rapid migration from local servers to Azure.

- **ETL & Orchestration:** Architected and deployed an automated ETL pipeline using Python and Dagster, replacing fragile manual SQL processes and ensuring reliable fee processing.
- **Business Intelligence:** Automated the data delivery lifecycle, integrating the pipeline to publish results directly to Power BI and auto-email PDF/Excel reports to collaborators.
- **Cloud Migration:** Successfully adapted the data architecture and orchestration logic to Azure during a rapid cloud transition.



Diagnostics Trainee

Boehringer Ingelheim — Vetmedica

2021 - 2022

Zapopan, MX

Bridged wet-lab operations and bioinformatics within an animal health laboratory, supporting viral genomic surveillance for veterinary vaccine development.

- **Genome sequencing:** Executed Illumina MiSeq sequencing and genome assembly protocols for viral ssRNA, building a strong understanding of sample processing in the wet lab.

Education



Ph.D. in Biomedical Sciences (Genomics Focus)

Centro de Ciencias Genómicas (CCG) — Universidad Nacional Autónoma de México (UNAM)

2025 - Present

Cuernavaca, MX



B.Sc. in Biotechnology Engineering

Instituto Tecnológico y de Educación Superior de Occidente (ITESO)

2018 - 2023

Guadalajara, MX

Technical Skills

Programming: Python, R, Bash, SQL

Cloud/Infra: DNAnexus, AWS, Docker, Linux

Genomics: Population & Statistical Genetics, Epidemiology, GWAS

AI Tools: Agentic coding tools like Antigravity CLI and Claude Code

Languages

Spanish: Native

English: Fluent (TOEFL iBT - C1)

Certifications



Bioinformatic Tools for High-Throughput Sequencing

USMBB - IBt UNAM

Jul. 2023



Introduction to Epidemiology

Nuffield Department of Public Health (NDPH) – University of Oxford 

ford

Jun. 2026

Engineering Biology Malice Analysis

Engineering Biology Consortium

Aug. 2020